AI ETHICS

Responsible Engineering Design and Innovation of AI Systems

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Thank you for the invitation to present in this course on AI Ethics: Global Perspectives. I'm Rafael Calvo, Chair of Engineering and Design at the Dyson School of Design and Engineering, Imperial College London. I'm also Imperial College's co-lead at the Liverhulme Center for the Future of Itelligence, and co-editor of the IEEE Transactions on Technology and Society. My research focuses on the design of intelligence systems that support wellbeing in areas of mental health, medicine, education, often applying techniques in machine learning, natural language processing, and data analytics. This has led me to publish about four books and about 200 papers on these topics. Now, I would like to start this talk by saying that in many ways, we have been here before. It's not all new. Engineers have built systems that follow the values of the time. For example, in the 18th and 19th centuries, engineers drove the first Industrial Revolution. This included exciting new materials, new forms of energy, and new technologies to increase productivity. These were the factories, steam engines and trains, rushing industry and people's lives into new frontiers. England was at the heart of this Industrial Revolution, taking the reigns with deep Victorian values. At the cusp of this Revolution, Prince Albert organized the Great Exhibition of 1851 - a Crystal Palace, an engineer wonder constructed entirely of glass and iron, rose from the ground by industrial magic. Visitors came from all over the world, including Charles Dickens, Lewis Carroll and Charles Darwin, to experience this industrial miracle. It was such a success that its revenues funded Albertopolis, and today Albertopolis is known as exhibition road, pictured here in modern-day London. The vision for Albertopolis was to celebrate and drive the advancement of industry in the UK. Today it is the home of Imperial College London, Royal College of Arts, the Victoria and Albert Museum - world leading institutions in design and engineering. But of course, there is much more to this success story. Taking a closer look, we find their impacts on humans and society that aren't quite so worthy. There are children working in these factories, workers are exploited, live in crowded, unsanitary conditions and disease is rampant. The air is filled with smoke on the water with waste. Here in London in 1850, we see death himself rowing through a toxic river Thames. My family is from Argentina, where many of its trains are literally the same as those that were installed by the English in the late 1800s. These trains allowed resources like cows and cereals to get from Argentina and Latin America to Europe. But the impacts on its Indigenous people and on the environment are still noticeable. What values were driving a technological progress that could overlook such massive costs and what things could have been done differently? While during the first Industrial Revolution our values were centered on improving the economy and expanding commerce and building the English Empire. This meant productivity and speed were the heroes of the day, engineers were expected to target this. The steam engines, electricity trains, planes, cars require that we take natural resources from the land, including water and air, and use them to feed the economy. In addition to the consumption of these resources, our products produce waste when we make them, when we use them, and when we dispose of them.

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One way to address this is to minimize the flow of resources from the natural environment. This is what we call the circular economy - waste gets fed back into industry. This changing thinking from a framework in which we rely on infinite depletion of pollution, of what actually if you need resources, to a framework in which the ideal innovation creates no waste it can't reuse reflects a major change in values. More of us now value our land, our water and air, we want innovations that take into account the environment and reduce its destruction. Changing values led to change in policy in the 1960s and 70s. And we now have regulation and professional processes to help ensure industry cannot go on ignoring environmental cost in this drive to commercial gain. Environmental impact assessments are now a standard practice in many types of engineering. It requires all stakeholders, all stakeholders to come together and agree on what in fact is acceptable, and how it can be minimized. Well, these are at least the ideas that stand to which this has been successful or hasn't been, can be felt in rising temperatures, weather catastrophes, deadly pollution. We continue to breathe in, increased rates of asthma, lung diseases, and our poorest have no clean drinking water. Although values have changed, much more needs to be done. But the first Industrial Revolution is now old news; today, you're living in the midst of a new revolution. And that's what we're discussing today here. Big Data, small data, the data about our lives, and our behavior is so large, so lucrative, that it now often is called the new oil. This makes companies like Facebook, Google and Amazon the new oil companies. This is a significant change in the way we envision society, technology and the economy. For many of today's technologies satisfying a human need is not the end goal, but only a means to an end. Facebook doesn't provide free social networking services to help the world, it has sought to make revenue. Meeting a human need for connectivity is only valuable insofar as it produces profit. And profit comes from the data generated and attention it attracts, and sells to advertisers. So what I want to talk today about is an approach to responsible engineering, design and innovation of systems, including AI, and we propose a human impact assessment for technology this year. I believe we can and we can describe a human impact assessment for technology that we call hired, that predicts and evaluates the impact the new digital technologies have on all stakeholders. These will acknowledge that these systems has psychosocial impacts on individuals and society, and that some of these may be negative for health, well-being and values - things like privacy, autonomy, and even democracy. We discuss what we can learn from these impact assessments in other fields, how to tap into existing ethical frameworks for AI, and the unique challenges associated with assessing impact on humans and society. Because of course, there are many differences between the environmental impact that we were talking first and these new one that has to do with AI and data. The environmental impact assessment of a factory is generally slow or static. Once installed, it has boundaries, it is regional, it happens where that factory or that bridge is created. It is largely anticipatory, we can predict the amount of pollution or noise that that factory will create; and it considers nature as the resource. Finally, we have a long history and that allows us to have research methods that we can use, make sure that we trust. On the other hand, when we are talking about the human impact of AI, we have systems that are dynamic, that self-learn, that are unbounded. The system that they developed today in London will have an impact in Argentina, in Latin America and Africa very soon after. Its continuous and interrogative. It's very hard to predict because it keeps changing in the way that users will interact with it. Maybe most importantly is that humans are the resource, like I said before, humans become a means to an end, rather than an end in itself. And finally, it's a set of new technologies that we don't yet know or understand well enough, so we don't have measures that people will generally trust. As we create technologies that promise to improve people's lives, engineers need to make similar ethical commitments to those that health professionals make with our patients. And that's one of the cases I want to make in this talk. These include supporting well-being, making sure they do no harm, supporting human autonomy, and being fair and just. Imagine what happens when doctors, hospitals, or others involved with your health are motivated only by profit. When you don't have NHS or national health system, you need to be very careful about the way that business models affect the health of nations. And that is why we now have this biomedical ethics framework that all in the industry must abide to. That is why we also have a strong regulatory framework, that limit how medical products are designed, manufactured, commercialized, and advertised. Our team has focused on psychological well-being as a key value, because we believe it's at the core of human experience with technology. This wish to be well is common to all sentient beings. But an evidence-based approach means we must define and understand well-being using the best tools and knowledge available. In our book Positive Computing, we review methods from psychology, neuroscience, and economics on what are the determinant factors of well-being and how they can be supported through technology design. Of these models, we have found one of the most effective within the technology context is self-determination theory that identifies autonomy, competence, and relatedness as the pillars of well-being or optimal functioning. Satisfaction of these psychological needs also predicts user satisfaction, engagement and motivation, making this theory especially applicable to technology experience. And it has like 40 years of empirical evidence from different cultures, different age groups, different business domains, from workplaces, to education, to health, and so forth. Competence here is about feeling capable and effective. In technology, this is very connected to our measures of usability and learnability. Autonomy is about our sense of willingness, and endorsement of actions and behaviors. It is not really about being independent, it is about endorsement, and an alignment with our values. Finally, relatedness refers to the feeling of being part of a community or having meaningful relationships with others. But the psychological needs by themselves are not enough. We have identified the need to be much more specific about where this needs satisfaction occurs which led us to develop these six spheres of technology experience, for which the yellow ones break down the user experience as we know it. The first sphere - adoption - refers to the decision-making experience between becoming aware of a new technology and acquiring it. Here we ask questions like To what extent is adoption of the technology autonomously motivated? Or to what extent does a potential user expect that technology will help them meet the need for competence or relatedness.

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Then we move into the yellow spheres which encapsulate what designers call the user experience. At the lowest granularity, this includes the interface which describes the experience of interacting with technology via its controls, navigation and functions during use. The quick questions to ask here are to what extent does direct interaction with technology via the user interface supports needs satisfaction. And this is an idea that good usability scores on user satisfaction. Step up, we look at the task level or how needs satisfying are the small tasks the technology supports, for example, in a fitness app, it might be counting steps - how needs satisfying is it to track your steps? Does it make you feel more or less competent? Can changes to how the tasks report is designed have impact on this? Tasks generally combine to make a larger activity or behavior, for example, exercising. So the question here is to what extent does the technology improve needs satisfaction with respect to exercise, which is the behavior that technology is ddesigned to support. Here, needs satisfaction should mediate engagement with a behavior, with the exercise in this case, in other domains, it will be mediated in other behavior, specific outcomes like health outcomes, learning outcomes, etc. And for the final sphere within this user experience, we have life, the life sphere looks at the individual overall experience, including everything beyond the technology, many technologies will not be expected to have an impact on these this life level. An egg timer, for example, could be measured, lower down this sphere, but we will hardly expected to have a profound impact on wellbeing in life. But in mindfulness up however, might certainly any technology that is keen to show it does not cause harm, for example, through indirect effects or high levels of addiction should be willing to measure the life level, because that is where sometimes the only place certain harms will be revealed. An easy targeted sample will be a technology that is highly addictive, will be highly needs satisfying at the interface and tasks levels, and that's why we are engaged. But if we overuse, it will begin to undermine autonomy and relatedness at the life level as relationships, for example, and values are impacted. Finally, we go beyond the user experience into the society sphere. This sphere includes direct and indirect impact on family, friends, and the non-users across society. While the idea of considering at this level is very, very daunting for many technology makers, engineers, like even myself, there is an increasing pressure to do so with more people concerned about technology ethics, and more people working on ways to make this possible, as all of you listening to this course. Now, I want to go a little bit more into the process. These spheres can help us tease apart the effects of technologies that are supporting psychological needs. In the metals paper, there is more detail about the use of ICT across these different spheres and impossible measures. But the process that we have also proposed brings these into a design approach that is common. And we have proposed a process for this responsible engineering design and innovation of AI technologies that uses the double time on design method that is very popular amongst design engineers and designers more generally. But it adds to these Double Diamond considerations on its impact on well-being and more generally on ethics. This happens when we are developing insights about the problem, when we hear, when we ideate possible solutions and when we evaluate the outcome, both in prototypes and then in a final deployment. Again, you can find more details about this approach in this paper here. But of course there are differences between AI and many other technology ethics problems. One of them is that because it requires such amounts of data and infrastructure, we feel the pressure to reuse its algorithms as much as possible. For many years I was a professor of software engineering, and reusability practice are at the core of our discipline. But when this happens, we put a significant focus on infrastructure. But then the thing is that the infrastructure gets used in many, what we call, vertical markets, or application domains. And a classification algorithm that we consider ethical when used, let's say, in medicine, may not be considered ethical in military applications. The same computer vision technique might be understood to be about safety in one situation, and about government control in another. But even when we think of the same industry, the way one company uses the algorithm might be consider appropriate, but the way it is used im another is not. In one, it has a positive impact, and in another, it might have a negative one. So which one do we regulate? Which one do we assess for impact? Many of the concerns we have today on AI ethics are conceived with this implicit understanding of hierarchy between core technologies or infrastructure, specific markets and clients. When we think about AI ethics, we often think of the AI infrastructure, the platforms that implement the algorithms that make, let's say, computer vision or natural language processing possible. For example, we ask these frameworks to be explainable. One key difference with environmental impact assessment is that the cost of the infrastructure for AI systems is much higher proportionally than that for the industrial projects. As for of course, we spend money in thinking how to create bridges. But most of the money goes into actually creating it. Creating each specific bridge or each specific factory. Occasionally, we think about the general application domain, for example, if we use a neural network for face recognition in security, it's not okay. But if we apply the same mathematical technique to detecting a tumor, it is okay. I think the impact of technology can only be understood when someone interacts with it. The meaning of AI is only realized when it is put in a specific technology, in a specific context, a specific place where certain people use it. This is the core of the interactionist approach, no? Interactional technology ethics conception. Of course, the three levels are important - infrastructure, market and client deployment. But the last one is the one we often forget. Again, this is an important question to understand which one will regulate on assessing the impact. So I'm going to use some two case studies that will hopefully help you understand my views on this more clearly. As mentioned before, the spheres can help us tease apart effects of technologies that are supporting psychological needs at one level, while undermining them at another. We can use these to assess the impact when we assess the impact on life in longer studies. One of the projects we worked on was head gear, the head gear project aimed at improving the mental health in the poor workplaces. It was funded by the Movember Foundation, and in partnership with ambulance, fire, and police services, these were male dominated industries. We've run workshops asking people about the problems they had. The designer then produced the science to support what we understood were users' values. We also understood the power relations between staff and employees. And the software architect is so we look for trusted distribution channels that were trusted. Of course, we also ask mental health researchers and we incorporated the best clinical practices into design. All our projects follow this participatory approach to design and I think this is critical for bringing ethics into software developing of AI systems. The evidence is that such approach is much more likely to lead to engagement in products, so it's good business. But also, I have over time become more aware of how they can also help us build products that promote health and are more ethical. In this case, we ran a randomized control trial, and it showed that it didn't have a negative impact and that it did support wellbeing. It showed that the technology had a positive effect, it reduced the risk of depression, and anxiety on the treatment group versus a control group. Of course, randomized control trials are not the solution to every problem. It works in health and some other domains, but not in every type of innovation. So my second example, is in a way more complicated and refers to understanding the impact on a wider group of stakeholders. And I want to bring here the importance of paying attention to the wider group of stakeholders. This was a project with reach out of Trylia, an organization that helps 1.8 million users, young people that are going through tough times. And one of the functionalities they have is a peer support group where young people can go and ask questions about relationships, or substance abuse, or the sort of problems that young people have. And one of the things that they have is a group of moderators that help respond and manage this online community. So we wanted to build a set of tools that help provide a better service to the young people going through tough times. And this was a moderator assistant. So we use natural language processing techniques to process the input, and then we'll use natural language generation techniques to generate template answers. So somebody will say: "Hello, everyone, I have been feeling awful for over four weeks, so my GP gave me Zoloft. Has anyone use it? Can you tell me about it?" So this is a mental health question. The system will diagnose this, could be depression because a person is taking an antidepressant. Thiswill move on to a classifier, and when the classifier says, and provides a diagnostic, we generated feedback questions related to diagnostic and any information that we may have about this particular user. So we run a pilot, well, we had a very detailed diagnostic manual that we use to train the algorithms and generate a data set that then we use to train the algorithm. And for the responses, we also run a trial. And technically the system was very successful, the accuracy of the diagnosing algorithm was much better than the baseline, we then run a pilot with 44 psychology students, and they'll automatically generate a test which thought to have been written by humans, by about 80% of the psychology students. Although the aim of this natural language generation was not to replace the human moderator, the system could potentially be very useful for providing moderators with draft responses, which basically will reduce the workload. And even if those responses require editing, we thought it would be beneficial to the moderators because it will allow them to deal with increased demand for feedback. But while this measures the system was satisfactory, one and a half years into the project, we realized that the project had failed. And we had to basically restart it from scratch. Although the automation had gone as planned, as we were doing or getting ready for the deployment, the industry partner came back and said he had to pull off. The system was putting off the moderators, moderators, were not going to engage and use it. So when we investigated with the moderators, we understood that they felt a loss of autonomy in the interaction. They perceived a long term loss of competences. They felt like a cog in a machine. And very wisely, I think the managers pulled the plug, and we had to go back to the drawing board. So we created a new version of the system - triage we called it - that listens to every new post as it comes in, and then makes a decision about how urgently it needs moderator's attention. Not a diagnosis, but the priorities that helps the moderator decide what to spend the time on. It's green if the post doesn't contain anything concerning, orange if it should be addressed at some point; red if it should be addressed as soon as they can. And we also then added the super red category, where if the user is talking about suicide, or self harm, or harming someone else. Reach out found this triage super useful, it actually helps. We added features to help moderators connect to each other to highlight the competencies they have, the skills. So for example, if an incoming post was about substance abuse, and one of the moderators, many of which have lived experiences, have experiences on substance abuse, somebody might recommend him to respond to the user. So it was helping moderators be better moderators, rather than replacing some of the tasks. The technical results were again, fantastic. When deployed, the system improved the process in all the key measures, you have 84 84% accuracy. The time to respond, that maybe was the most important, was reduced by 80% for crisis post. And this is the most critical one because they talk about suicides, so you need to respond very quickly. And maybe most importantly, the moderators have been engaged with the system for about 40 years now. In summary, I think AI ethics is technology ethics. Technologies mediate our experiences of the outside world, our schools, our health systems, our workplaces. They mediate how we make decisions. But the meaning of the technology can only become evident during our interaction with it. It cannot be understood by itself. AI can seem sometimes successful in some measures, but then have a negative impact in others. Exploring then all stakeholders and their values during the design phase, is important. Regulation or just evaluation of the infrastructure is not enough. Actual deployments need to be assessed. Thank you very much.